

Autosomal Dominant Polycystic Kidney Disease

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DEFINITION

Autosomal dominant polycystic kidney disease (ADPKD) is a multi-system disorder characterized by multiple, bilateral kidney cysts and associated with cysts in other organs, such as the liver, pancreas, and arachnoid membranes.¹ It is a genetic disorder caused mainly by mutations in either of two major genes and inherited in an autosomal dominant pattern, with variable expression. Although benign (nongenetic) kidney cysts are common with aging, an underlying inherited disease should be suspected in patients with multiple bilateral kidney cysts, even if the kidney function is normal.

ETIOLOGY AND PATHOGENESIS

The common ADPKD proteins, polycystin-1 and polycystin-2, play a critical role in the normal function of the primary cilium that is essential to maintaining the differentiated phenotype of tubular epithelium.²

Genetic Mechanisms

ADPKD is genetically heterogeneous with two common genes identified (Fig. 46.1): *PKD1* (chromosome 16p13.3) and *PKD2* (4q21). Recently, three additional genes have been identified in pedigrees with an atypical ADPKD presentation: *GANAB*,³ *DNAJB11*,⁴ and *ALG9*⁵; the encoded proteins are involved in the maturation and processing of polycystins. Autosomal dominant polycystic liver disease (ADPLD) also exists as an independent entity and is genetically heterogeneous; the first two genes identified (*PRKCSH* in chromosome 19 and *SEC63* in chromosome 6) account for about one-third of isolated ADPLD cases. More recently, mutations to *GANAB*, *ALG8*, *SEC61B*, and *LRP5* have also been shown to cause ADPLD.⁶

Evidence from animal models of ADPKD and analysis of cystic epithelia have shown that kidney cysts may develop from loss of functional polycystin with somatic inactivation of the normal allele. However, cysts can develop even if the protein is not completely lost, as demonstrated by animal models expressing incompletely penetrant alleles.⁷

Polycystic Kidney Disease Proteins

Polycystin-1 (PC1; PKD1 protein) and polycystin-2 (PC2; PKD2 protein) belong to a subfamily of transient receptor potential (TRP) channels. PC1 (~440 kDa) has the structure of a receptor or adhesion molecule and contains a large extracellular N region, 11 transmembrane regions, and a short intracellular C region (see Fig. 46.1). PC1

interacts with PC2 through a coiled-coil domain in the C-terminal portion and with multiple other proteins at different extracellular and intracellular sites. PC1 is found in the primary cilia, plasma membrane at focal adhesions, desmosomes, adherens junctions, and possibly endoplasmic reticulum (ER) and nuclei. PC2 (TRPP2; ~110 kDa) contains a short N-terminal cytoplasmic region, six transmembrane domains, and a short C-terminal portion. PC2 is localized predominantly to the ER but also to the plasma membrane, primary cilium, centrosome, and mitotic spindles in dividing cells.² A cryo-EM structure indicates 1 PC1 to 3 PC2 molecules.⁸ PC1 and PC2 are also found at high concentrations in exosomes, which are shed into the urine and physically interact with primary cilia, possibly exerting an effect on other cells in the nephron.

Mechanisms of Cyst Formation

Experimental data indicate that the timing of ciliary loss or *Pkd1* inactivation determines the rate of development of cystic disease. Inactivation in the developing kidney results in rapid progression.⁹ Interestingly, the loss of *Pkd1* and cilia results in less severe polycystic kidney disease (PKD) than loss of *Pkd1* alone.¹⁰

The polycystins are involved in the detection of extracellular cues at primary cilia, cell-cell contacts, and cell-matrix contacts and are essential to maintain the differentiated phenotype of the tubular epithelium. Reduction in one of the polycystins to below a critical threshold results in inability to maintain planar polarity, increased rates of proliferation, expression of a secretory phenotype, and remodeling of the extracellular matrix.² PC1 and PC2 in the primary cilium may be required for the increase in cytosolic calcium that occurs in response to ciliary bending.¹¹ PC2 is a TRP channel (TRPP2) and functions as a calcium release channel in the ER.¹² PC1 interacts with and modulates the maturation and function of PC2, and vice versa.¹³ PC1 and PC2 also interact with additional calcium channel proteins. Precisely how intracellular calcium homeostasis is altered in ADPKD remains uncertain, but many studies show reduced resting intracellular calcium, ER calcium stores, and store-operated calcium entry in primary cell cultures or microdissected samples from human and rodent polycystic tissues² (Fig. 46.2).

A common finding in animal models of PKD is increased levels of cyclic adenosine monophosphate (cAMP), not only in the kidney but also in the liver and vascular smooth muscle.¹⁴ Tissue levels of cAMP are determined by the activities of membrane-bound and soluble adenyl cyclases and cAMP phosphodiesterases (PDEs), themselves subject to complex regulatory mechanisms. Reduced intracellular calcium

Polycystins

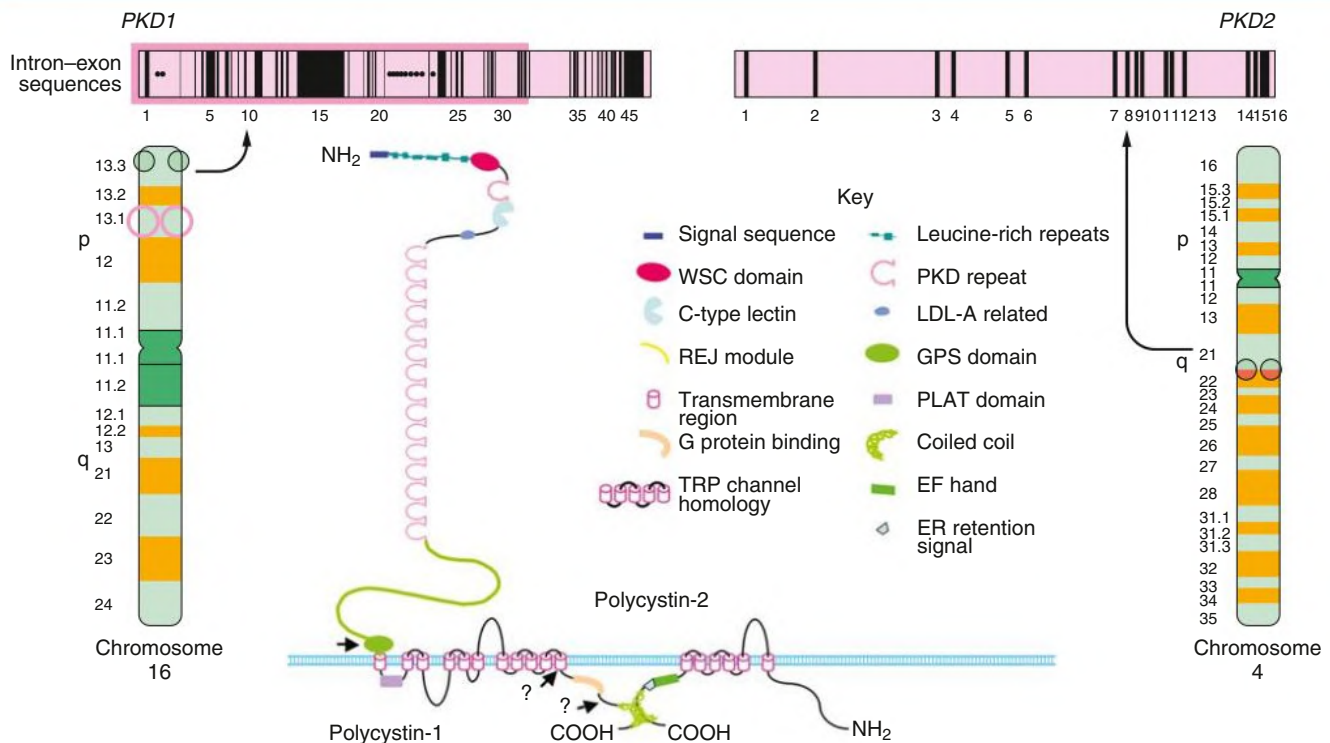


Fig. 46.1 Polycystins: Genes, Messenger RNAs, and Proteins. Diagrammatic representation of chromosome 16 (left) and chromosome 4 (right). Intron-exon sequences of *PKD1* (upper left) and *PKD2* (upper right). Diagram of proposed structural features of the polycystin-1 and polycystin-2 proteins (center). ER, Endoplasmic reticulum; GPS, G-protein-coupled receptor proteolytic site; LDL, low-density lipoprotein; PKD, polycystic kidney disease; PLAT, polycystin-1, lipoxygenase, alpha-toxin; REJ, receptor for egg jelly; TRP, transient receptor potential; WSC, wall integrity and stress response component.

in PKD may activate calcium inhibitable AC6 or AC5, directly inhibit calcium/calmodulin-dependent PDE1, and indirectly inhibit cGMP inhibitable PDE3, thereby accounting for the accumulation of cAMP and activation of protein kinase A (PKA), which, in turn, contributes to the development and progression of PKD by stimulating cystic fibrosis transmembrane conductance regulator (CFTR)-driven chloride and fluid secretion and cell proliferation (see Fig. 46.2).

Chloride enters across basolateral $\text{Na}^+\text{-K}^+\text{-2Cl}^-$ cotransporters, driven by the sodium gradient generated by basolateral $\text{Na}^+\text{-K}^+\text{-ATPase}$, and exits across apical PKA-stimulated CFTR. Basolateral recycling of potassium occurs through the KCa3.1 channel.

cAMP exerts opposite effects on cell proliferation in different cell types. cAMP and PKA signaling enhance several proliferative pathways (extracellular signal-regulated kinase [ERK]) in cells derived from polycystic kidneys and inhibit proliferation in cells derived from normal human kidney cortex.^{15,16} Treatment of normal human kidney or murine collecting duct cells with calcium channel blockers replicates the proliferative response of the ADPKD cells to cAMP, thus linking this response to the reduction in intracellular calcium that results from disrupting the polycystin pathway.¹⁷ Conversely, treatment of ADPKD cyst-derived cells with calcium channel activators or calcium ionophores restores the normal antimitogenic response to cAMP (see Fig. 46.2).

Liver Cyst Development

Liver cysts arise by excessive proliferation and dilation of biliary ductules and peribiliary glands. Estrogen receptors, insulin-like growth factor 1

(IGF-1), IGF-1 receptors, and growth hormone receptors are expressed in the epithelium lining the hepatic cysts, and estrogens and IGF-1 stimulate hepatic cyst-derived cell proliferation. Cyst growth is also promoted by growth factors and cytokines secreted into the cyst fluid.

Hypertension

Hypertension is a major clinical manifestation and predictor of outcome in ADPKD (see Clinical Manifestations). Several factors contribute to the development of hypertension in ADPKD. Activation of the intrarenal renin-angiotensin system (RAS) likely plays an important role, but whether the circulating RAS is inappropriately activated is controversial. The expression of PC1 and PC2 in vascular smooth muscle and endothelium, along with enhanced vascular smooth muscle contractility and impaired endothelium-dependent vasorelaxation in ADPKD, suggest that disruption of polycystin function directly contributes to hypertension. Other factors include increased sympathetic nerve activity and plasma endothelin-1 levels and insulin resistance.¹

Endothelial vasodilation and constitutive nitric oxide synthase activity are reduced in subcutaneous resistance vessels from patients with ADPKD and normal glomerular filtration rate (GFR). Flow-induced vasodilation of the brachial artery is inconsistently impaired, whereas pulse wave reflection is amplified, suggesting a predominant involvement of small resistance vessels. Reduced coronary flow velocity reserve and increased carotid intima-media thickness in normotensive patients with normal GFR suggest that atherosclerosis starts early in the course of ADPKD.

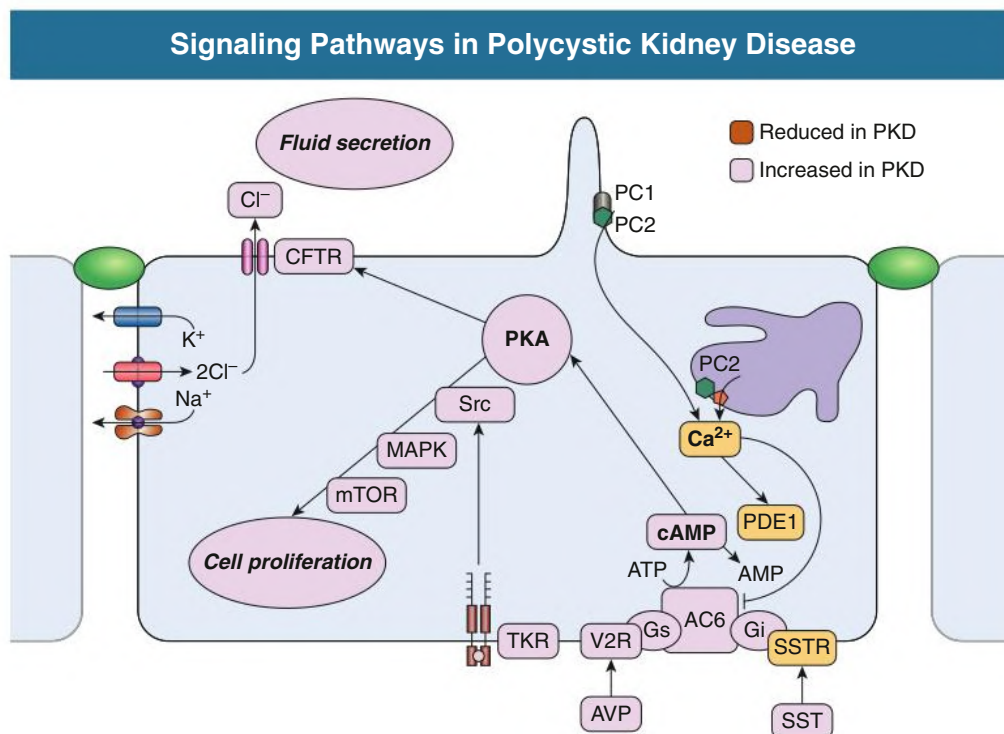


Fig. 46.2 Signaling Pathways in Polycystic Kidney Disease (PKD). Pathways that are upregulated or downregulated in PKD and the rationale for potential therapies. Dysregulation of intracellular calcium homeostasis leads to intracellular accumulation of cyclic adenosine monophosphate (cAMP), activation of protein kinase A (PKA), cystic fibrosis transmembrane conductance regulator (CFTR) phosphorylation, and stimulation of chloride-driven fluid secretion. In the setting of reduced intracellular calcium, PKA activates Src, mitogen-activated protein kinase (MAPK)/extracellular signal-regulated kinase (ERK), and mammalian target of rapamycin (mTOR) signaling. Activation of tyrosine kinase receptors (TKR) for several growth factors contributes to the activation of Src and downstream proliferative pathways. Therapies currently under clinical investigation target G protein-coupled receptors (modulating activity of adenylyl cyclase 6 [AC6] and generation of cAMP), Src, mTOR, and TKRs. AVP, Arginine vasopressin; PDE1, phosphodiesterase-1; SST, somatostatin; SSTR, somatostatin receptor.

Reduced nitric oxide endothelium-dependent vasorelaxation in ADPKD may be caused by increased plasma levels of asymmetric dimethylarginine, a mechanism common to all hypertension associated with kidney disease.

Altered polycystin function in cardiac fibroblasts likely accounts for the increased frequency of valvular heart disease in ADPKD.

EPIDEMIOLOGY

ADPKD occurs worldwide and in all races, with a prevalence of genetically affected individuals at birth estimated at 1 in 400 to 1 in 1000. In most patients the diagnosis is made decades later, and some patients are never diagnosed. Therefore, at any point in time, only a fraction of genetically affected individuals are aware of having the disease. Clinical registry data suggest point prevalence rates of diagnosed cases ranging from 1 in 543 to 1 in 4000. The proportion of end-stage kidney disease (ESKD) caused by ADPKD is less among Blacks than among Whites because of a higher incidence of other causes of ESKD. Yearly incidence rates for ESKD caused by ADPKD in males and females, respectively, are 8.7 and 6.9 per 1 million (1998–2001, United States), 7.8 and 6.0 per million (1998 and 1999, Europe), and 5.6 and 4.0 per million (1999 and 2000, Japan). Age-adjusted sex ratios greater than unity (1.2–1.3) and recent genotype/phenotype studies suggest more progressive disease in males than in females.^{18–20} In recent studies, the age of onset of ESKD has increased in both sexes, and all-cause

mortality has decreased, possibly because of improved detection and control of hypertension.¹

PHENOTYPIC VARIABILITY

Genic, allelic, and gene-modifier effects contribute to the high phenotypic variability of ADPKD. *PKD1*-associated disease is more severe than *PKD2*-associated disease (age at ESKD, 58 years vs. 79 years for *PKD1* and *PKD2*, respectively).²¹ The greater severity of *PKD1* is caused by development of more cysts at an early age, not faster cyst growth.¹ Both *PKD1* and *PKD2* can be associated with severe polycystic liver disease (PLD) and vascular abnormalities.²² Because of the lesser severity of the kidney involvement, the prevalence of *PKD2*-associated disease has likely been underestimated in clinical studies.

Mutations in *PKD1* and *PKD2* are highly variable and often “private” (unique to a kindred). The ADPKD Mutation Database (<http://pkdb.mayo.edu>) lists 1621 likely pathogenic *PKD1* mutations identified in 2417 families with a total of 2905 variants, including silent polymorphisms. Also, 256 likely pathogenic *PKD2* mutations are listed in 553 families, with a total of 358 different variants. A much smaller number of patients with mutations to *GANAB*, *DNAJB11*, and *ALG9* have been described.³

Allelic factors have an effect on the severity of ADPKD. Recent studies in large cohorts have shown that the type of *PKD1* mutation, but not its position, correlates strongly with kidney survival. The median

age at onset of ESKD was 55 years for carriers of a truncating mutation and 67 years for carriers of a nontruncating mutation.²¹ Hypomorphic or incompletely penetrant *PKD1* or *PKD2* alleles have been described. These alleles alone may result in mild cystic disease; two such alleles cause typical to severe disease and, in combination with an inactivating allele, may be associated with early-onset disease that mimics autosomal recessive PKD (ARPKD).¹ A prognostic score (PRO-PKD score) based on the gene mutated and type of mutation helps predict kidney outcomes and enable the personalization of therapeutic management in patients with ADPKD.^{2,23}

The large intrafamilial variability of ADPKD highlights a role for genetic background in disease presentation. Age at onset of clinical manifestations in ADPKD is less variable within than between families, which suggests a common familial modifying background for early and severe disease expression (e.g., mutations or variants in genes encoding other cystoproteins). The contiguous deletion of the adjacent *PKD1* and *TSC2* is characterized by childhood PKD with additional clinical signs of tuberous sclerosis complex. Other modifying loci are likely to account for more common and subtle intrafamilial variability.

DIAGNOSIS

Only individuals who have been properly informed about the advantages and disadvantages of screening should be offered presymptomatic screening. If ADPKD is diagnosed, the patient should receive appropriate genetic counseling, and risk factors such as hypertension can be identified and treated early. If ADPKD is absent, the patient can be reassured. Disadvantages of presymptomatic screening relate to insurability and employability. Presymptomatic screening of children is not recommended until more effective therapy for the disease becomes available.

Kidney Imaging

Kidney ultrasound is used for presymptomatic testing because of cost and safety. Revised criteria have been proposed to improve the diagnostic performance of ultrasound in ADPKD (Table 46.1). At least three (unilateral or bilateral) kidney cysts are sufficient for diagnosis of at-risk individuals 15 to 39 years of age; two cysts in each kidney are sufficient for diagnosis for ages 40 to 59 years.²⁴ For at-risk individuals age 60 and older, four or more cysts in each kidney are required.

The specificity and positive predictive value (PPV) of ultrasound are close to 100% using these criteria, but their sensitivity and negative predictive value (NPV) when applied to PKD2 patients age 15 to 59 years are low. This is a problem in the evaluation of potential kidney donors, in which exclusion of the diagnosis is important. Different criteria have therefore been proposed to exclude a diagnosis of ADPKD in an individual at risk from a family with an unknown genotype. An ultrasound finding of normal kidneys or one kidney cyst in an individual aged 40 years or older has an NPV of 100%. The absence of any kidney cysts provides near certainty that ADPKD is absent in at-risk individuals aged 30 to 39 years, with a false-negative rate of 0.7% and NPV of 98.7%. A normal or indeterminate ultrasound scan does not exclude ADPKD with certainty in an at-risk individual younger than 30 years. A study of 73 affected and 82 nonaffected individuals suggested that finding fewer than five cysts by magnetic resonance imaging (MRI) is sufficient to exclude the diagnosis of ADPKD in potential living related kidney donors.²⁵ Contrast-enhanced computed tomography (CT) scanning with thin slices likely provides similar information, but this has not been proven.

Genetic Testing

Genetic testing can be performed when a precise diagnosis is needed and the results of imaging are indeterminate, although a case can now be

TABLE 46.1 Ultrasound Criteria for the Diagnosis of Autosomal Dominant Polycystic Kidney Disease

Age (y)	Criteria	PPV	NPV
Original Ravine's PKD1 Diagnostic Criteria			
15–29	≥2 cysts, unilateral or bilateral	99	88
30–39	≥2 cysts in each kidney	100	88
40–59	≥2 cysts in each kidney	100	95
≥60	≥4 cysts in each kidney	100	100
Revised Unified Diagnostic Criteria			
15–29	≥3 cysts, unilateral or bilateral	100	86
30–39	≥3 cysts, unilateral or bilateral	100	86
40–59	≥2 cysts in each kidney	100	95
≥60	≥4 cysts in each kidney	100	100
Revised Diagnostic Criteria (when diagnosis needs to be excluded)			
15–29	≥1 cyst	97	91
30–39	≥1 cyst	94	98
40–59	≥2 cysts	97	100
≥60	≥3 cysts in each kidney	100	100

NPV, Negative predictive value; PPV, positive predictive value.

made for broader screening to obtain a firm diagnosis. Molecular diagnosis of ADPKD is complicated by duplication of the *PKD1* gene, which means it may not be efficiently screened by whole-exome sequencing. For diagnostic purposes, next-generation sequencing panels of PKD genes are now generally employed that allow for higher throughput screening and lower costs.²⁶ Molecular testing by direct DNA sequencing is now informative in at least 90% of patients. However, because many mutations are unique and up to one-third of *PKD1* changes are missense, the pathogenicity of some changes is difficult to prove. De novo mutations and mosaicism also can complicate interpretation of results.²⁷

In preimplantation genetic diagnosis (PGD), genetic analysis is performed on single blastomeres from preimplantation embryo biopsy specimens obtained after in vitro fertilization (IVF), and only embryos unaffected by the disease are selected for transfer. PGD for ADPKD is complicated by the genetic heterogeneity of the disease and the large size and complex structure of the *PKD1* gene but has been performed for ADPKD. PGD should be included in the discussion of reproductive choices with patients with ADPKD, but it is only available in certain countries and the acceptance of this technique is influenced by personal values and by the severity of the disease.¹

DIFFERENTIAL DIAGNOSIS

Kidney cysts can be a manifestation of many other systemic diseases. Conditions to consider when presentation is not typical of ADPKD include ARPKD, tuberous sclerosis complex, von Hippel–Lindau disease, kidney cysts and diabetes from *HNF1B* mutations, and oro-facio-digital syndrome type I, as well as medullary sponge kidney and simple kidney cysts. These are discussed further, including differential diagnosis, in Chapter 47. If the patient has ESKD, acquired cystic disease also should be considered (see Chapter 92).

CLINICAL MANIFESTATIONS

ADPKD is a multisystem disorder. Multiple kidney and extrarenal manifestations of ADPKD can cause significant complications.

BOX 46.1 Kidney Manifestations of Autosomal Dominant Polycystic Kidney Disease

Functional Manifestations

- Concentrating defect
- Reduced kidney blood flow

Hypertension → Target Organ Damage

- Cardiac
- Cerebrovascular
- Arteriolosclerosis and glomerulosclerosis
- Peripheral vascular disease

Pain, Caused by

- Cyst hemorrhage
- Gross hematuria
- Nephrolithiasis
- Infection
- Kidney enlargement

Reduced Glomerular Filtration Rate, Possibly Caused by

- Interstitial inflammation
- Apoptosis of tubular epithelial cells
- Hypertensive glomerulosclerosis
- Compression atrophy

Kidney Manifestations

A number of clinical features that result from kidney damage can be identified (Box 46.1). Reduction in urinary concentrating capacity and glomerular hyperfiltration are early functional abnormalities that can be observed in some children and adolescents with ADPKD.

Kidney Size

Kidney size increases with age, and kidney enlargement eventually occurs in 100% of patients with ADPKD. The severity of the structural abnormality correlates with the manifestations of ADPKD, such as pain, hematuria, hypertension, and kidney impairment.¹ Massive kidney enlargement can lead to compression of local structures, resulting in such complications as inferior vena cava (IVC) compression and digestive symptoms. Most manifestations are directly related to the development and enlargement of kidney cysts. The Consortium for Radiologic Imaging Studies of Polycystic Kidney Disease (CRISP), a prospective study of 241 patients by annual MRI, has shown that total kidney volume (TKV) and cyst volumes increased exponentially.²⁸ Rates of growth were relatively constant, averaging 5.3% per year, but were highly variable from patient to patient. Baseline TKV predicted the subsequent rate of increase in kidney volume and decline in GFR.²⁹ An imaging classification of ADPKD based on height-adjusted TKV and age may facilitate the selection of patients with rapidly progressive disease (class 1C–E) for enrollment into clinical trials and for disease-modifying treatments.³⁰

Pain

Episodes of acute kidney pain are seen frequently; causes include cyst hemorrhage, infection, stone, and, rarely, tumor, and these must be investigated thoroughly. A few patients with ADPKD with kidney enlargement and structural distortion develop chronic flank pain without an identifiable cause.

Hematuria and Cyst Hemorrhage

Visible hematuria may be the initial presenting symptom and occurs in up to 40% of patients with ADPKD over the course of the disease. Many have recurrent episodes. Differential diagnosis includes cyst hemorrhage, stone, infection, and tumor. Cyst hemorrhage is a frequent complication and produces gross hematuria when the cyst communicates with the collecting system. Frequently, the cyst does not communicate with the collecting system, and flank pain without hematuria occurs. It can manifest with fever, raising the possibility of cyst infection. On occasion, a hemorrhagic cyst will rupture, resulting in a retroperitoneal bleed that can be significant, potentially requiring transfusion. In most patients, cyst hemorrhage is self-limited, resolving within 2 to 7 days. If symptoms of hematuria or flank pain last longer than 1 week or if the initial episode of hematuria occurs after age 50 years, neoplasm should be excluded.

Urinary Tract Infection and Cyst Infection

Urinary tract infection (UTI) is common in ADPKD, but its incidence may have been overestimated because sterile pyuria is common in these patients. UTI presents as cystitis, acute pyelonephritis, cyst infection, and perinephric abscesses. As in the general population, females are affected more frequently than males. Most infections are caused by *Escherichia coli*, *Klebsiella* and *Proteus* species, and other Enterobacteriaceae. The route of infection in acute pyelonephritis and cyst infection is usually retrograde from the bladder; therefore cystitis should be promptly treated to prevent complicated infections.

Both CT and MRI are sensitive to detect complicated cysts and can provide anatomic definition, but the findings are not specific for infection. Nuclear imaging, especially indium-labeled white blood cell scanning, is useful, but false-negative and false-positive results are possible. F-labeled fluorodeoxyglucose (FDG) positron emission tomography (PET) has recently been used for detection of infected cysts.¹ FDG is taken up by inflammatory cells because of their high metabolic rate but is filtered by the kidneys, is not reabsorbed, and appears in the collecting system, which may limit its use in diagnosis of kidney cyst infections; its present role is for diagnosis of infected liver cysts. FDG-PET is expensive and not widely available, but it provides rapid imaging with high spatial resolution, high target-to-background ratio, low radiation burden, and high interobserver agreement.

When there is fever and flank pain with suggestive diagnostic imaging but blood and urine cultures are negative, cysts should be aspirated under ultrasound or CT guidance to culture the organism and inform the selection of antimicrobial therapy.

Nephrolithiasis

Kidney stone disease occurs in about 20% of patients with ADPKD. Most stones are composed of uric acid, calcium oxalate, or both. Uric acid stones are more common in ADPKD than in stone formers without ADPKD. Urinary stasis secondary to distorted kidney anatomy may play a role in the pathogenesis of nephrolithiasis. Predisposing metabolic factors include decreased ammonia excretion, low urinary pH, and low urinary citrate concentration.

Stones can be difficult to diagnose on imaging in ADPKD because of cyst wall and parenchymal calcification. The distorted anatomy can cause difficulty in localizing stones to the collecting system on plain radiographs. Intravenous (IV) urography has the advantage of specifically localizing stone material to the collecting system and may provide clues to stone composition. IV urography also can detect precalyceal tubular ectasia, found in 15% of patients with ADPKD. CT urography has replaced IV urography in many centers; it is more sensitive in detecting small or radiolucent stones and for differentiating stones

Effect of Blood Pressure on Kidney Survival in Autosomal Dominant Polycystic Kidney Disease

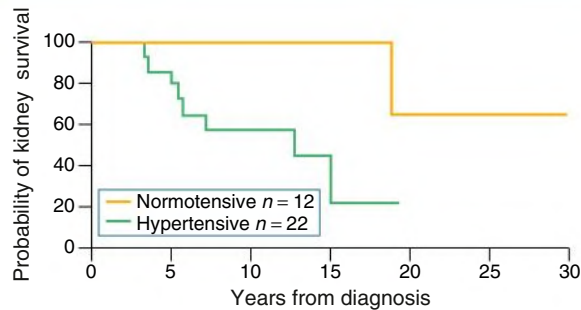


Fig. 46.3 Patients with polycystic kidney disease and hypertension at diagnosis have less probability of kidney survival than those with normal blood pressure.

from tumor, clot, and cyst wall or parenchymal calcification. Dual-energy CT is increasingly used to distinguish between calcium and uric acid stones.

Hypertension

Hypertension is the most common manifestation of ADPKD and a major contributor to kidney disease progression and cardiovascular morbidity and mortality (Fig. 46.3). Albuminuria and hematuria are more common in hypertensive patients with ADPKD and correlate with the rate of kidney function loss. Hypertension also may increase morbidity from valvular heart disease and intracranial aneurysms, which are common in ADPKD.

Ambulatory blood pressure (BP) monitoring of children or young adults without diagnosed hypertension often reveals elevated BP, attenuated nocturnal BP dipping, and exaggerated BP response during exercise. A study stratified 65 children by BP into three cohorts: hypertensive (≥ 95 th percentile), borderline hypertensive (75th to 95th percentile), and normotensive (≤ 75 th percentile).³¹ Both the hypertensive and the borderline hypertensive children had significantly higher left ventricular (LV) mass indices than the normotensive children. Among normotensive children, indices were significantly higher in those within the upper quartile of normal BP. These observations suggest that target organ damage develops early in ADPKD and that antihypertensive treatment may be indicated in children with ADPKD and borderline hypertension.

End-Stage Kidney Disease

In most patients, kidney function is maintained within the normal range, despite relentless growth of cysts, until the fourth to sixth decade of life. By the time kidney function starts declining, the kidneys usually are greatly enlarged and distorted with little recognizable parenchyma on imaging studies. At this stage, the average rate of GFR decline is 4.4 to 5.9 mL/min per year. Nevertheless, ESKD is not inevitable in ADPKD. Up to 77% of patients are alive with preserved GFR at age 50 years and 52% at age 73 years. Males tend to progress to kidney failure more rapidly and require kidney replacement therapy at a younger age than females. Other risk factors for kidney failure include Black race, diagnosis of ADPKD before age 30 years, first episode of hematuria before age 30 years, onset of hypertension before age 35 years, hyperlipidemia, low level of high-density lipoprotein cholesterol, sickle cell trait, and *PKD1* truncating mutations. Recently, a post hoc analysis of the HALT PKD clinical trials showed an association of dietary sodium with the rate of kidney growth in

patients with ADPKD with estimated GFR (eGFR) >60 mL/min/1.73 m² and with the rate of decline in kidney function in those with eGFR between 25 and 60 mL/min/1.73 m².³²

Several mechanisms account for loss of GFR. The CRISP study confirmed that kidney and cyst volumes are the strongest predictors of declining GFR²⁹ and also found that kidney blood flow (or vascular resistance) is an independent predictor.³³ The latter suggests that vascular remodeling may account for cases in which the decline of kidney function seems disproportionate to the severity of the cystic disease, particularly in patients reaching ESKD in the seventh decade of life or later.³⁴ Other factors, such as heavy use of analgesics, may contribute to chronic kidney disease (CKD) progression in some patients.

Extrarenal Manifestations

Polycystic Liver Disease

PLD is the most common extrarenal manifestation of ADPKD. PLD is associated with both *PKD1* and *PKD2* genotypes. In contrast to the kidney phenotype, the ADPKD genotype is not associated with the severity or growth rate of PLD in patients with ADPKD.²² In addition, PLD also occurs as a genetically distinct disease in the absence of kidney cysts (ADPLD). Most simple hepatic cysts are solitary, and PLD should be suspected when four or more cysts are present in the hepatic parenchyma. The liver in PLD contains multiple microscopic or macroscopic cysts that result in hepatomegaly (Fig. 46.4), but typically there is preservation of normal hepatic parenchyma and liver function.

Hepatic cysts are exceedingly rare in children with ADPKD. Their frequency increases with age and may have been underestimated by ultrasound and CT studies. Their prevalence by MRI in the CRISP study was 58%, 85%, and 94%, respectively, in participants age 15 to 24, 25 to 34, and 35 to 46 years. Females develop more cysts at an earlier age than males. Females who have multiple pregnancies or who have used oral contraceptives (OCs) or estrogen replacement therapy (ERT) in the postmenopausal period may have worse disease. After menopause, the volume of polycystic livers often remains stable or may even decrease.²²

Typically, PLD is asymptomatic, but reported symptoms have become more frequent as the life span of patients with ADPKD is prolonged with dialysis and transplantation. Symptoms result from mass effect or from complications related to the cysts themselves.³⁵ Symptoms include dyspnea, orthopnea, early satiety, gastroesophageal reflux, mechanical low back pain, uterine prolapse, and even rib fracture. Other complications caused directly by mass effect include hepatic venous outflow obstruction, IVC compression, portal vein compression, and bile duct compression presenting as obstructive jaundice. Hepatic venous outflow obstruction is an uncommon condition caused by severe extrinsic compression of the intrahepatic IVC and hepatic veins by cysts, rarely with superimposed thrombosis. Symptomatic cyst complications include cyst hemorrhage, which occurs less frequently than kidney cyst hemorrhage, cyst infection, and the rare occurrence of torsion or rupture of cysts. Hepatic cyst infection can be a serious complication and typically presents with localized pain, fever, leukocytosis, elevated sedimentation rate, and often elevated alkaline phosphatase. Enterobacteriaceae are the most common microorganisms causing cyst infection. The same imaging techniques discussed for the investigation of kidney cyst infections may be useful for the localization of infected cysts in the liver.

Congenital hepatic fibrosis, always found in association with ADPKD, can rarely coexist with ADPKD. Contrary to PKD, which affects members of several generations in these families, congenital hepatic fibrosis is seen in only one generation and is not transmitted vertically, suggesting the importance of modifier genes. These patients present with manifestations of portal hypertension, but hepatocellular function is normal.

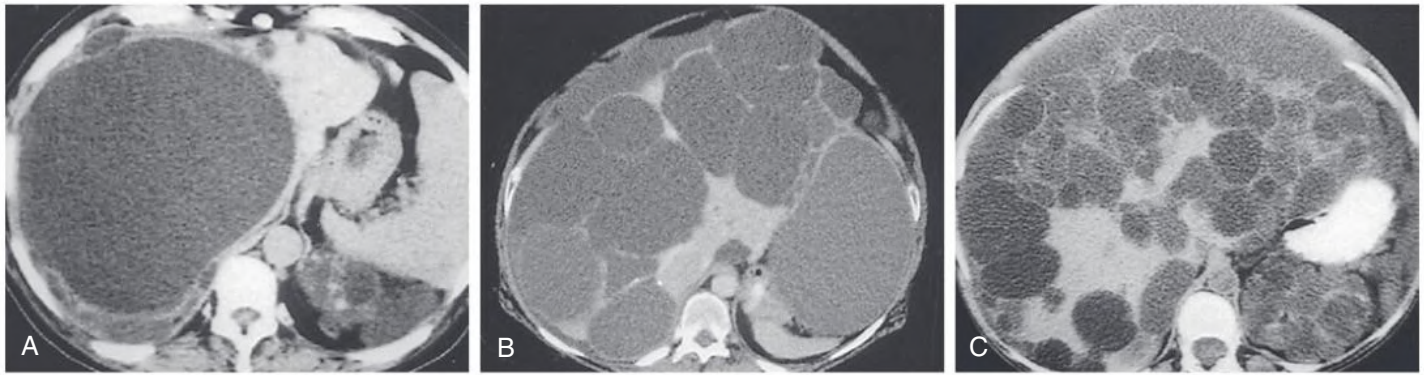


Fig. 46.4 Variable Presentation of Symptomatic Polycystic Liver Disease. (A) Hepatomegaly caused by a very large, isolated, dominant cyst. (B) Hepatomegaly caused by several large cysts. (C) Hepatomegaly caused by multiple smaller cysts throughout the hepatic parenchyma.

Clinical Manifestations and Classification of Intracranial Aneurysms

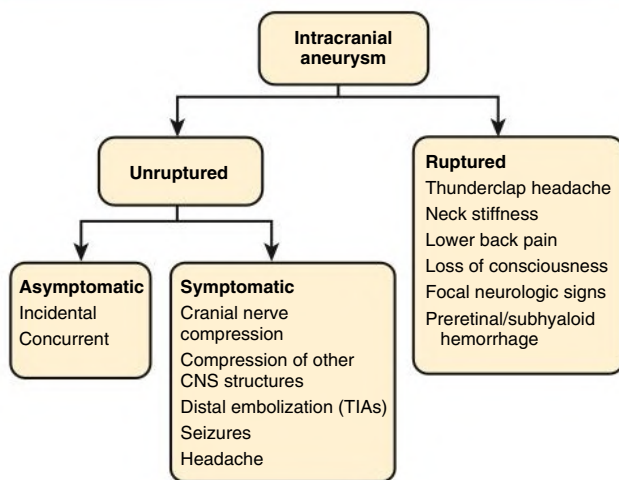


Fig. 46.5 Clinical manifestations and classification of intracranial aneurysms. CNS, Central nervous system; TIAs, transient ischemic attacks.

Intracranial Aneurysms

Intracranial aneurysms occur in about 8% of patients with ADPKD. There is some familial clustering; intracranial aneurysms occur in 6% of patients with a negative family history and 16% of those with a positive family history. Most are asymptomatic. Focal findings, such as cranial nerve palsy and seizure, may result from compression of local structures by an enlarging aneurysm (Fig. 46.5). Yearly rupture rates increase with size, ranging from less than 0.5% for aneurysms <5 mm in diameter to 4% for aneurysms >10 mm. Rupture carries a 35% to 55% risk for combined severe morbidity and mortality. The mean age at rupture in ADPKD is 39 years (vs. 51 years in the general population), with a range of 15 to 69 years. Most patients have normal kidney function, and up to 29% have normal BP at rupture.

Although still controversial, the largest follow-up study of asymptomatic, unruptured intracranial aneurysms discovered by presymptomatic imaging and a review of the literature suggest that widespread screening is not indicated for all patients with ADPKD because most intracranial aneurysms found by presymptomatic imaging are small, have a low risk for rupture, and require no treatment because the risks of intervention exceed any risk for rupture.³⁶ Indications for targeted imaging in patients with a good

life expectancy include family history of intracranial aneurysm or subarachnoid hemorrhage, previous aneurysmal rupture, preparation for elective surgery with potential hemodynamic instability, high-risk occupations (e.g., airline pilots), and significant patient anxiety despite adequate information about the risks. Magnetic resonance angiography is the modality of choice for presymptomatic imaging because it is noninvasive and does not require IV contrast administration. CT angiography is a satisfactory alternative if there is no contraindication to IV contrast. Repeated imaging of patients with a strong family history of intracranial aneurysms or aneurysmal rupture after 5 to 10 years is recommended.

Other Vascular Abnormalities

In addition to intracranial aneurysms, ADPKD is associated with other vascular abnormalities, such as thoracic aortic and cervicocephalic arterial dissections, coronary artery aneurysms, and retinal artery and vein occlusions (Fig. 46.6). Thoracic aortic dissection is seven times more common in ADPKD than in the general population in autopsy series, but reported cases are rare. Rare patients with coronary aneurysms can present with cardiac ischemia and thrombus in the aneurysm in the absence of atherosclerotic disease. Several case reports describe abdominal aortic aneurysms in ADPKD. However, a prospective ultrasound study showed neither a wider aortic diameter nor a higher prevalence of abdominal aortic aneurysms in patients with ADPKD compared with an unaffected kindred in any age group.

Valvular Heart Disease and Other Cardiac Manifestations

Mitral valve prolapse is the most common valvular abnormality and has been demonstrated in up to 25% of patients with ADPKD by echocardiography. Mitral regurgitation, tricuspid regurgitation, and tricuspid prolapse also occur more frequently in ADPKD than in unaffected kindreds. Aortic regurgitation may be associated with dilation of the aortic root. On histologic examination, valvular tissue shows myxoid degeneration with disruption of collagen, as seen in Marfan and Ehlers-Danlos syndromes. Although the lesions may progress with time, they rarely require valve replacement. Echocardiography is not indicated unless a murmur is detected on physical examination. Coexistence of ADPKD and idiopathic cardiomyopathies (dilated, hypertrophic, and LV noncompaction) appears to be higher than expected by chance.³⁷ Small, hemodynamically insignificant pericardial effusion can be detected by CT scanning in up to 35% of patients with ADPKD.

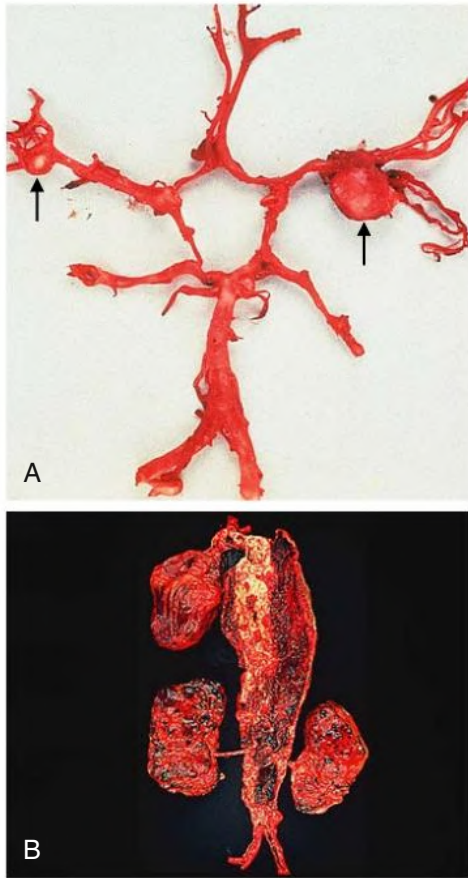


Fig. 46.6 Vascular Manifestations of Autosomal Dominant Polycystic Kidney Disease (ADPKD). (A) Gross specimen demonstrating bilateral aneurysms of the middle cerebral arteries (*arrows*). (B) Gross specimen demonstrating a thoracic aortic dissection extending into the abdominal aorta in a patient with ADPKD.

Other Associated Conditions

Cyst formation has been described in such diverse organs as the pancreas, seminal vesicles, and arachnoid membrane (Fig. 46.7). Seminal vesicle cysts, usually multiple and bilateral, are found in 40% of ADPKD compared with 2% of nonaffected males. Ovarian cysts are not associated with ADPKD. Pancreas and arachnoid membrane cysts are present in 5% and 8% of patients, respectively. Pancreatic cysts are almost always asymptomatic, with rare occurrences of recurrent pancreatitis and possibly chance associations of intraductal papillary mucinous tumor or carcinoma. Epididymal and prostate cysts also may occur with increased frequency. Sperm abnormalities with defective motility are common in ADPKD and rarely may be a cause of male infertility. Spinal meningeal diverticula may occur with increased frequency and rarely manifest with intracranial hypotension (orthostatic headache, diplopia, hearing loss, ataxia) caused by cerebrospinal fluid leak. The prevalence of colonic and duodenal diverticula also may be increased.

PATHOLOGY

Polycystic kidneys are diffusely cystic and enlarged (Fig. 46.8). Size varies from normal to weighing more than 4 kg. The outer and cut surfaces show numerous spherical cysts of varying size, which are distributed evenly between cortex and medulla. The epithelium lining the cysts is characterized by hyperplastic changes, including flat nonpolypoid hyperplasia, polypoid hyperplasia, and microscopic adenomas

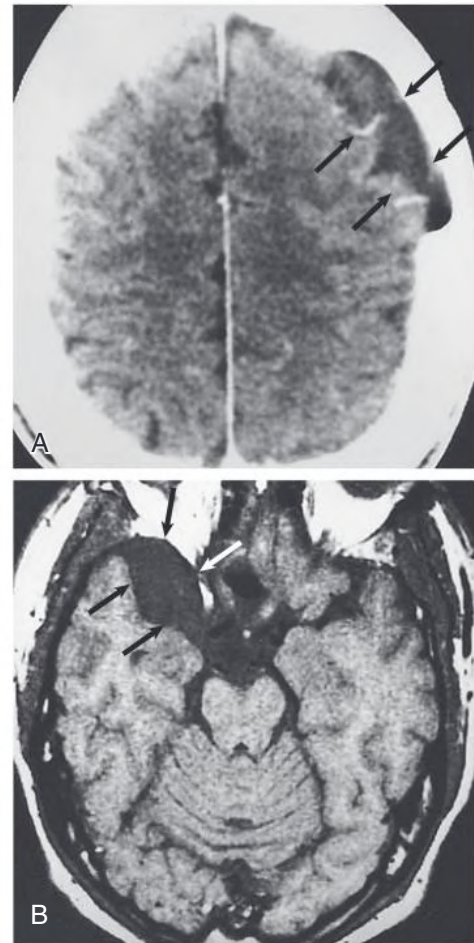


Fig. 46.7 Extracranial Manifestations of Autosomal Dominant Polycystic Kidney Disease (ADPKD). Computed tomography (A) and magnetic resonance imaging (B) scans demonstrate cysts in the arachnoid membrane (*arrows*) in ADPKD.



Fig. 46.8 Polycystic Kidneys. Greatly enlarged polycystic kidneys from a patient with autosomal dominant polycystic disease compared with a normal kidney (*middle*).

(Fig. 46.9), as well as increased rates of cell proliferation and apoptosis. Despite the frequency of hyperplastic lesions and microscopic adenomas, the incidence of renal cell carcinoma is not increased.

Cysts arise from all segments of the nephron and collecting ducts. As they grow, cysts dissociate from the parent tubule and eventually become isolated, fluid-filled sacs. There is no agreement on whether

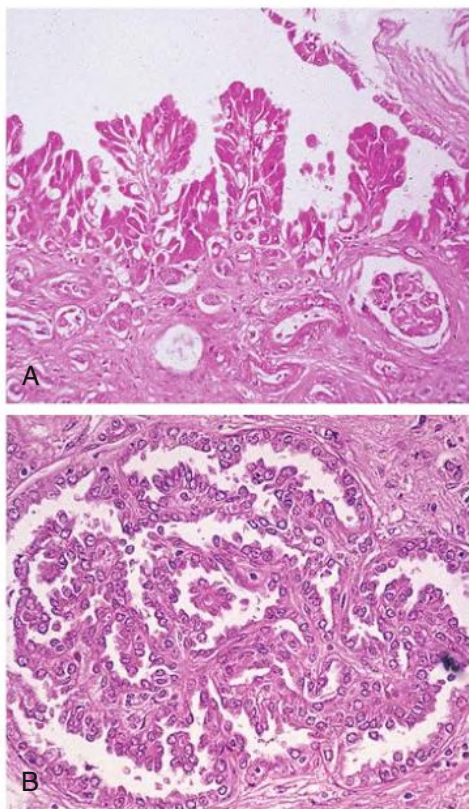


Fig. 46.9 Kidney Cyst Histology in Autosomal Dominant Polycystic Kidney Disease (ADPKD). (A) Papillary hyperplasia of cyst epithelium. (B) Papillary tubular adenoma in kidney of patient with ADPKD. (Magnification $\times 200$.)

the cysts originate preferentially from particular tubular segments. Most studies indicate that cysts are predominantly of distal nephron and collecting duct origin. Studies in advanced kidney disease showing proximal tubular cysts may be confounded by the effects of obstruction and acquired kidney cystic disease.

Polycystic kidneys demonstrate advanced sclerosis of preglomerular vessels, interstitial fibrosis, and tubular epithelial hyperplasia, even in patients with normal GFR or early kidney failure. Sclerosis involves both afferent arterioles and interlobular arteries. Interstitial fibrosis is also prominent, even in early disease. It is associated with an interstitial infiltrate of macrophages and lymphocytes.

TREATMENT

Therapy is directed toward the kidney and extrarenal complications of ADPKD in an effort to limit morbidity and mortality. Advances in the understanding of the genetics of ADPKD and mechanisms of cyst development and growth have raised hopes for treatments specifically directed toward limiting the development and progression of the disease, and some of these treatments are now being evaluated in clinical trials, with one treatment now approved (see Novel Therapies).

Flank Pain

Causes of flank pain that may require intervention, such as infection, stone, and tumor, should be excluded. Care should be taken to avoid long-term administration of nephrotoxic agents, such as combination analgesics and nonsteroidal anti-inflammatory drugs. Narcotic analgesics should be reserved for the management of acute pain. Patients with chronic kidney pain are at risk for narcotic and analgesic dependence,

and a psychological evaluation and a supportive attitude by the physician are essential. Reassurance, lifestyle modification, and avoidance of aggravating activities may be helpful. Tricyclic antidepressants are helpful as in other chronic pain syndromes, with a generally well-tolerated side effect profile. Splanchnic nerve blockade with local anesthesia or corticosteroids results in pain relief prolonged beyond the duration of the local anesthetic.

When distortion of the kidneys by large cysts is deemed responsible for the pain and conservative measures fail, cyst decompression should be considered. Cyst aspiration, under ultrasound or CT guidance, is a relatively simple procedure. To prevent the reaccumulation of cyst fluid, sclerosing agents such as 95% ethanol, acidic solutions of minocycline, sodium tetradecyl sulfate, and, more recently, sodium tetradecyl sulfate mixed with room air (foam sclerotherapy),³⁸ may be used. Minor complications include microhematuria, localized pain, transient fever, and systemic absorption of the alcohol. More serious complications, such as pneumothorax, perirenal hematoma, arteriovenous fistula, urinoma, and infection, are rare. Complications from aspiration of centrally located cysts are more common, and the morbidity of the procedure is proportional to the number of cysts treated.

If multiple cysts are contributing to pain, laparoscopic or surgical cyst fenestration may be of benefit. Surgical decompression is effective in 80% to 90% of patients at 1 year, and 62% to 77% have sustained pain relief for over 2 years (Fig. 46.10A). Surgical intervention does not accelerate the decline in kidney function but does not appear to preserve declining GFR either (see Fig. 46.10B). Laparoscopic fenestration is as effective as open surgical fenestration in short-term follow-up in patients with limited disease, and there is a shorter, less complicated recovery period compared with open surgery. Previous abdominal surgery with possible adhesion formation is a relative contraindication.

Other interventions are available for the management of pain in ADPKD whose roles have not been fully defined. Laparoscopic kidney denervation has been used in combination with cyst fenestration and may be considered, particularly in polycystic kidneys without large cysts. Thoracoscopic sympathectomy is effective in some patients but has significant morbidity. Catheter-based kidney denervation has been reported to be successful in a few patients, but a prospective protocolized evaluation of this approach is needed. Laparoscopic and retroperitoneoscopic nephrectomy and arterial embolization have been used to treat symptomatic polycystic kidneys in patients with ESKD.

Cyst Hemorrhage

Episodes of cyst hemorrhage are self-limited, and patients respond well to conservative management with bed rest, analgesics, and increased fluid intake to prevent obstructing clots. Rarely, bleeding is more severe, with extensive subcapsular or retroperitoneal hematoma causing significant decrease in hematocrit and hemodynamic instability. This requires hospitalization and transfusion. The antifibrinolytic agent tranexamic acid has been successfully used in some patients, but no controlled studies have been performed.³⁹ The dose should be reduced at lower levels of eGFR. Potential adverse effects of tranexamic acid therapy include glomerular thrombosis and ureteral obstruction from clots. In patients with unusually severe or persistent hemorrhage, segmental arterial embolization can be successful. If not, surgery may be required to control bleeding.

Urinary Tract and Cyst Infection

Because most kidney cyst infections begin as cystitis, prompt treatment of symptomatic cystitis and asymptomatic bacteriuria is indicated to prevent retrograde seeding of the kidney parenchyma. Antibiotics that require glomerular filtration, such as highly polar aminoglycosides, are

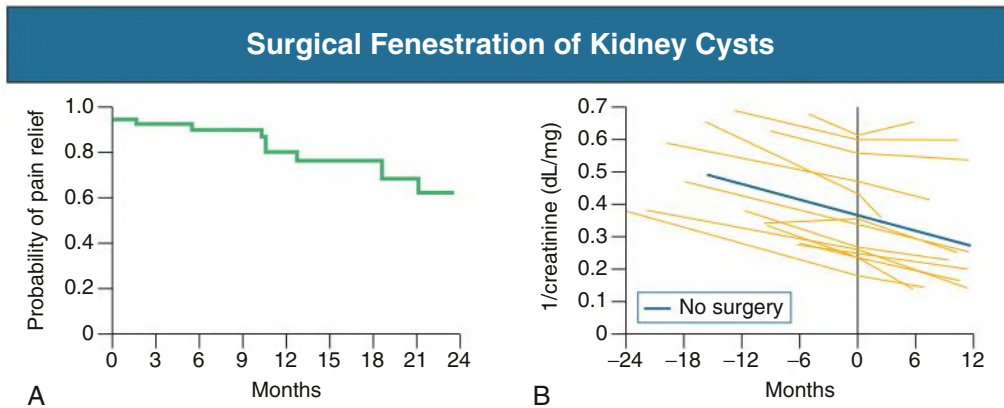


Fig. 46.10 Surgical Cyst Fenestration for Symptomatic Autosomal Dominant Polycystic Kidney Disease. (A) Effects on relief of pain. (B) Rate of decline of kidney function. Orange lines indicate the course of kidney function in individual patients who underwent cyst fenestration at month 0. The green line represents the mean rate of decline for all patients in the study.

not effective for upper UTI in severe kidney impairment. Cyst infection is often difficult to treat despite prolonged therapy with an antibiotic to which the organism is susceptible. Treatment failure occurs if antibiotics do not penetrate the cyst epithelium and achieve therapeutic concentrations within the cysts. Lipophilic agents have been shown to penetrate cysts reliably and have an acid dissociation constant (pK_a) that allows for favorable electrochemical gradients into acidic cyst fluid. Therapeutic agents of choice include trimethoprim-sulfamethoxazole and fluoroquinolones, both of which have favorable intracystic therapeutic concentration gradients at physiologic pH in gradient and nongradient cysts.

If fever persists after 1 to 2 weeks of appropriate antimicrobial therapy, infected cysts should be drained percutaneously or surgically. In the case of end-stage polycystic kidneys, nephrectomy should be considered. If fever recurs after stopping antibiotics, complicating features such as obstruction, perinephric abscess, and stone should be excluded. If no such complicating features are identified, the antibiotic course should be extended and may require several months to fully eradicate infection.

Nephrolithiasis

Treatment of nephrolithiasis in patients with ADPKD is the same as that in patients without ADPKD (see Chapter 60). Potassium citrate is the treatment of choice in the three stone-forming conditions associated with ADPKD: uric acid lithiasis, hypocitraturic calcium oxalate nephrolithiasis, and distal acidification defects. Extracorporeal shock wave lithotripsy and percutaneous nephrostolithotomy are reported to be 82% and 80% successful, respectively, without increased complications compared with patients without ADPKD.¹ Flexible ureterorenoscopy with laser fragmentation also has been used safely and effectively and cannot cause traumatic nephron loss.¹

Hypertension

Control of hypertension is essential because uncontrolled hypertension accelerates the decline in kidney function and aggravates extrarenal complications. Antihypertensive agents of choice and optimal BP targets in ADPKD have not been established. Angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs) increase kidney blood flow in ADPKD, have a favorable side effect profile, and may have renoprotective properties that go beyond BP control. The HALT PKD study, a recent double-blind, placebo-controlled randomized clinical trial (RCT), examined the role of RAS blockade in both early and advanced ADPKD.^{40,41} Monotherapy with ACE

inhibitors gave good BP control in the majority of patients. In early ADPKD (15–49 years of age, $GFR > 60$ mL/min/1.73 m²), rigorous BP control (target range 95–110/60–75 mm Hg) was associated with slower increase in TKV, faster eGFR decline during the first 4 months of treatment followed by a slower eGFR decline thereafter without an overall eGFR effect, a lesser increase in kidney vascular resistance, and a greater decline in LV mass index, after 8 years of follow-up.⁴² Patients with more severe ADPKD, determined by age-adjusted kidney volume, were more likely to benefit from rigorous BP control.⁴² Dual RAS blockade with lisinopril and telmisartan had no beneficial effect compared with lisinopril alone in these patients, nor in more advanced ADPKD (aged 18–64 years with GFR between 25 and 60 mL/min/1.73 m²). The HALT PKD study therefore supports the use of an ACE inhibitor or ARB as the antihypertensive medication of choice in most patients with ADPKD and a rigorous BP target (95–110/60–75 mm Hg) in young patients with ADPKD with normal kidney function and without other significant comorbidities.

Progressive Kidney Failure

General strategies to delay progression of CKD are discussed in Chapter 82. Hypertension plays an important role in the progression of ADPKD to ESKD. In addition to the HALT PKD clinical trials, long-term follow-up of participants in the Modification of Diet in Kidney Disease (MDRD) study suggested that individuals with ADPKD randomized to a low BP target (mean arterial pressure [MAP] < 92 mm Hg) experienced significantly less ESKD and combined ESKD/death than those randomized to the usual BP target (MAP < 107 mm Hg).¹

Preclinical studies, and evidence suggesting the increased vasopressin effect on the kidney and cAMP levels are involved in cyst progression, have led to the ingestion of supplemental water sufficient to achieve a urinary osmolality below 250 mOsm/kg H₂O (~3 L in most patients) being recommended for patients with ADPKD with $eGFR > 30$ mL/min, although no RCTs support this approach.⁴³ Exclusions would include severe protein or sodium restriction, reduced effective intravascular volume, taking diuretics or drugs enhancing the release of arginine vasopressin (AVP), or abnormal voiding. Serum sodium concentration should be monitored.

Patients with ADPKD have reduced morbidity and mortality on dialysis compared with patients with ESKD from other causes. Females appear to do better than males. The good outcome in ADPKD may result from higher endogenous erythropoietin production, better maintenance of hemoglobin, or lower comorbidity. Rarely, hemodialysis can be complicated by intradialytic hypotension if there is IVC

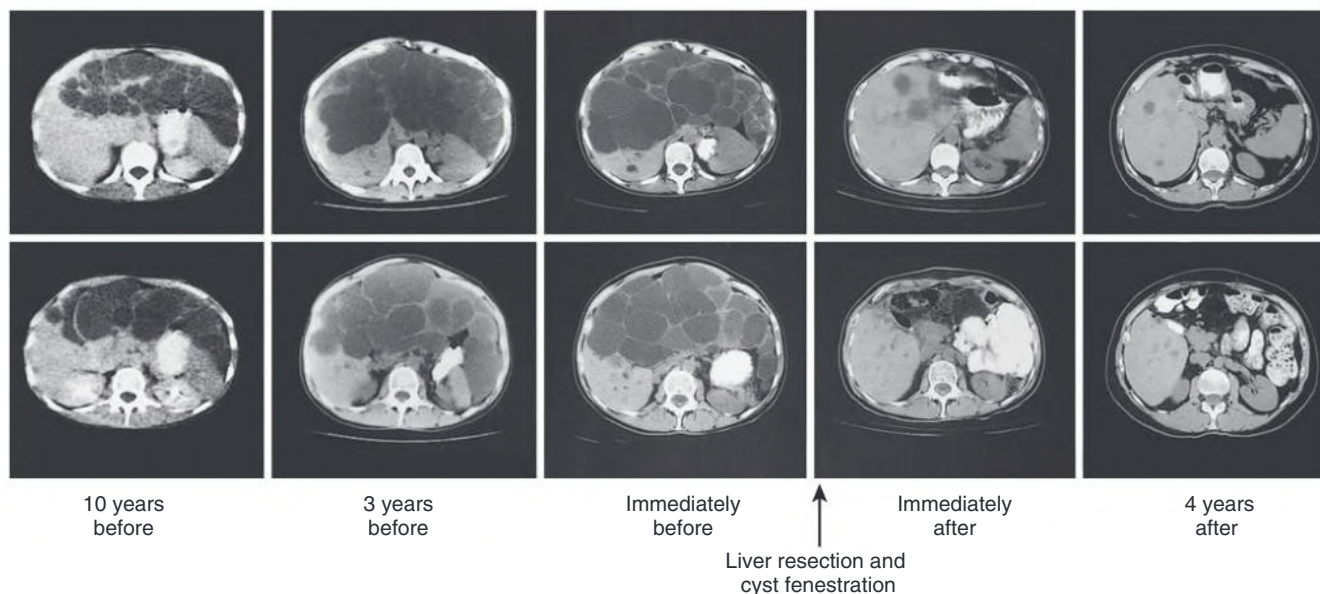


Fig. 46.11 Hepatic Resection in Autosomal Dominant Polycystic Kidney Disease (ADPKD). Computed tomography scans of the abdomen from two patients with ADPKD obtained are 10 years before (column 1), 3 years before (column 2), immediately before (column 3), immediately after (column 4), and 4 years after (column 5) liver resection and cyst fenestration, demonstrating long-term, sustained reduction in liver size after the procedure.

compression by a medially located kidney cyst. Despite kidney size, peritoneal dialysis can usually be performed in ADPKD, although there is increased risk for inguinal and umbilical hernias, which require surgical repair.

Polycystic Liver Disease

Usually asymptomatic, PLD requires no treatment. When symptomatic, therapy is directed toward reducing cyst volume and hepatic size. Noninvasive measures include avoiding excessive use of ethanol, other hepatotoxins, and possibly cAMP agonists (e.g., excessive caffeine), which have been shown to stimulate cyst fluid secretion in vitro. Histamine-2 blockers and somatostatin may reduce secretion of secretin and secretory activity of cyst walls. Estrogens are likely to contribute to cyst growth, but the use of OCs and postmenopausal ERT are contraindicated only if the liver is significantly enlarged and the risk for further hepatic cyst growth outweighs the benefits of estrogen therapy. Rarely, symptomatic PLD may require invasive measures to reduce cyst volume and hepatic size. Options include percutaneous cyst aspiration and sclerosis, laparoscopic fenestration, and open surgical fenestration. Cyst aspiration is the procedure of choice if symptoms are caused by one or a few dominant cysts or by cysts that are easily accessible to percutaneous intervention. To prevent the reaccumulation of cyst fluid, sclerosis with minocycline, 95% ethanol, or foam sclerotherapy with sodium tetradecyl sulfate is often successful. Laparoscopic fenestration can be considered for large cysts that are more likely to recur after ethanol sclerosis, or if several cysts are present that would require multiple percutaneous passes to be treated adequately. Partial hepatectomy with cyst fenestration is an option because PLD often spares a part of the liver with adequate preservation of hepatic parenchyma and liver function⁴⁴ (Fig. 46.11). In the rare case in which no segments are spared, liver transplantation may be necessary. When no better options are available, long-acting somatostatin analogs can be effective in reducing, and in some cases, reversing the growth of polycystic livers (see under Novel Therapies).

When a hepatic cyst infection is suspected, any cyst with unusual appearance on an imaging study should be aspirated for diagnostic purposes. The best management is percutaneous cyst drainage in combination with antibiotic therapy. Long-term oral antibiotic suppression or prophylaxis should be reserved for relapsing or recurrent cases. As for kidney cyst infection, antibiotics of choice are trimethoprim-sulfamethoxazole and the fluoroquinolones.

Intracranial Aneurysm

Ruptured or symptomatic intracranial aneurysm requires surgical clipping of the neck of the aneurysm. Asymptomatic aneurysms measuring <5 mm, diagnosed by presymptomatic screening, can be observed with repeated magnetic resonance angiography at 6 months, then annually and less frequently after stability of the aneurysm has been established. If the size increases, surgery is indicated. Definitive management of aneurysms between 6 and 9 mm remains controversial. Surgical intervention is usually indicated for all unruptured aneurysms that are at least 10 mm in diameter. For patients with high surgical risk or technically difficult lesions, endovascular treatment with detachable platinum coils may be indicated.⁴⁵

NOVEL THERAPIES

A better understanding of the pathophysiology and the availability of animal models have facilitated the development of promising candidate drugs for clinical trials (see Fig. 46.2). Of the candidates, only tolvaptan has so far been approved by regulatory agencies for the treatment of rapidly progressive ADPKD⁴⁶ and has entered clinical practice.

Vasopressin Antagonists

The effect of vasopressin, through V_2 receptors, on cAMP levels in the collecting duct, the major site of cyst development in ADPKD, and the role of cAMP in cystogenesis provided the rationale for successful preclinical trials of vasopressin V_2 receptor antagonists.² High

water intake by itself also exerted a protective effect on the development of PKD in a rat model of PKD (PCK rat), probably because of suppression of vasopressin. Genetic elimination of AVP in these rats yielded animals born with normal kidneys that remained relatively free of cysts unless an exogenous V₂ receptor agonist was administered.⁴⁷

The Tolvaptan Efficacy and Safety in Management of Autosomal Dominant Polycystic Kidney Disease and its Outcomes (TEMPO) 3:4 trial, a phase III, double-blind, 3-year RCT of a split daily dose of tolvaptan in 1445 patients with rapidly progressive disease and relatively preserved GFR (18–50 years old, estimated creatinine clearance > 60 mL/min, TKV > 750 mL), showed a reduction in the rates of kidney growth (from 5.5% to 2.8% per year) and of eGFR decline (from 3.70 to 2.72 mL/min/1.73 m² per year), and in the frequency of ADPKD-related adverse events (kidney pain, hematuria, UTI).⁴⁸ However, tolvaptan was associated with a higher frequency of aquaresis-related adverse events (polyuria, thirst, urinary frequency, nocturia) that led to drug discontinuation in 8.3% of tolvaptan-treated patients. In addition, clinically significant increases in liver enzymes (>2.5 times the upper limit of normal) were seen in 4.9% of tolvaptan-treated versus 1.2% of placebo-treated patients, and two patients had severe hepatocellular toxicity. Small increases in serum sodium and uric acid levels were also seen more frequently in the patients taking tolvaptan. The Replicating Evidence of Preserved Renal function: an Investigation of Tolvaptan Safety and Efficacy (REPRISE) trial, a phase III, randomized withdrawal, double-blind, 1-year trial, in 1370 18- to 55-year-old or 56- to 65-year-old patients with eGFR of 25 to 65 or 25 to 44 mL/min per 1.73 m², respectively, showed a reduction in the rate of eGFR decline (from 3.61 to 2.34 mL/min/1.73 m²).⁴⁹ Reversible elevations in the alanine aminotransferase of more than three times the upper limit of normal occurred in 5.6% in the tolvaptan group and 1.2% in the placebo group. No patients had severe hepatocellular toxicity, likely because of more frequent monitoring and earlier discontinuation of tolvaptan. An analysis of 97 patients continuously treated with tolvaptan at a single center for 4.6 (range, 1.1–11.2) years showed that the effect on the rate of eGFR decline was sustained and cumulative over time.⁵⁰

Tolvaptan is approved for the treatment of rapidly progressive ADPKD in the United States, Japan, Canada, the European Union, Switzerland, Norway, South Korea, and Australia. Criteria of rapid progression vary widely among countries, using a combination of age, baseline GFR, GFR slope, baseline TKV, and TKV rate of growth. Height-adjusted TKV (HtTKV) adjusted by age (image classification) is a practical and validated method to identify risk of rapid progression (Class 1C–1E) at early stages of the disease when the GFR is still normal.⁵¹ Because of the potential hepatocellular toxicity, the Food and Drug Administration (FDA) in the United States requires a risk evaluation and mitigation strategy (REMS) program, including liver function testing before initiation and at specific intervals (after 2 and 4 weeks, then monthly for 18 months, and every 3 months thereafter) after initiation of treatment. A clinical trial of lixivaptan, a vasopressin V₂ receptor antagonist predicted by a quantitative systems toxicology mathematical model to have a lower risk of hepatotoxicity compared with tolvaptan, is ongoing.

Somatostatin Analogs

Somatostatin acting on somatostatin receptors inhibits cAMP accumulation not only in the kidney but also in the liver. Somatostatin has a half-life of approximately 3 minutes, so more stable synthetic peptides (octreotide, lanreotide, pasireotide) have been developed for clinical use, which vary in stability and receptor affinity. In preclinical

studies, these drugs reduced cAMP levels and proliferation of cholangiocytes in vitro, expansion of liver cysts in three-dimensional collagen culture, and development of kidney and liver cysts and fibrosis in animal models orthologous to ADPKD and ADPKD. Three initial small RCTs of octreotide or lanreotide^{52–54,55–57} showed similar results. Kidney growth was halted during the first year of treatment and then resumed, possibly at a lower rate than without treatment. Liver volume decreased by 4% to 6% during the first year of treatment, and this reduction was sustained during the second year. Similar results were obtained in a larger RCT of patients (A Long-Acting Somatostatin on Disease Progression in Nephropathy Due to Autosomal Dominant Polycystic Kidney Disease [ALADIN]), in which a trend toward stabilization of GFR was observed after the first year.⁵⁸ The results of three additional RCTs have been recently reported. The Developing Interventions to Halt Progression of Autosomal Dominant Polycystic Kidney Disease-1 (DIPAK-1) trial, an open-label, 2.5-year RCT of lanreotide-LAR, 120 mg intramuscularly every 28 days, in 309 patients with eGFR 30 to 60 mL/min/1.73 m², confirmed a significant reduction in liver and kidney cyst growth, but no attenuation of eGFR decline.⁵⁹ ALADIN-2, a double-blind, placebo-controlled 3-year RCT of octreotide-LAR, 40 mg intramuscularly every 28 days, in 100 patients with eGFR between 15 and 40 mL/min/1.73 m² showed a significant effect on kidney growth and no effect on the rate of eGFR decline but a significant reduction in the rate of progression to ESKD, particularly in the patients with most advanced disease.⁶⁰ A double-blind, 1-year RCT of pasireotide (60 mg subcutaneously every 28 days) in 48 patients aged older than 18 years, with eGFR greater than 30 mL/min/1.73 m² and liver volume greater than 4000 mL, showed a reduction in liver volume and total kidney growth rates but no effect on eGFR.⁶¹

Octreotide and lanreotide are generally well tolerated. Self-resolving abdominal cramps and loose stools are common in the first few days after the injections. Other adverse effects include injection site granuloma and pain, cholelithiasis, steatorrhea, weight loss, and, rarely, hair loss. In DIPAK-1, an increased frequency of liver cyst infection in patients with a past history of a similar episode was observed. A high incidence of diabetes and hyperglycemia was observed in patients treated with pasireotide.

Mammalian Target of Rapamycin Inhibitors

The mammalian target of rapamycin (mTOR) is activated in animal models of PKD. Patients with the contiguous *PKD1/TSC2* gene syndrome have more severe PKD than those with ADPKD alone. This observation suggests a convergence of signaling pathways downstream from PC1 and the TSC proteins tuberlin and hamartin that control the activity of mTOR. Studies in rodent models of PKD showed that the mTOR inhibitors sirolimus and everolimus significantly prevent cyst expansion and protect kidney function. However, two large RCTs using everolimus and sirolimus for 18 to 24 months have shown no benefit.^{62,63}

Other Investigational Therapies

Pravastatin, an HMG-CoA reductase inhibitor, slowed the rate of kidney growth in a small RCT in children and young adults with ADPKD.⁶⁴ A phase II, multicenter, double-blind RCT of the Src inhibitor bosutinib reduced the rate of kidney growth, with a trend to worsen GFR (ClinicalTrials.gov NCT01233869). Clinical trials of the glycosylceramide synthase inhibitor venglustat, Nrf2 activator bardoxolone, vasopressin V₂ receptor antagonist lixivaptan, multikinase inhibitor KD019, metformin, pioglitazone, nicotinamide, and triptolide are ongoing (see Fig. 46.2).

TRANSPLANTATION

Transplantation is the treatment of choice for ESKD in patients with ADPKD. There is no difference in patient or graft survival between patients with ADPKD and other ESKD populations. Living donor transplants also have graft survival no different from that of non-ADPKD populations. However, living related transplantation has only recently been widely practiced in the ADPKD population. In 1999, 30% of kidney transplants for patients with ADPKD were from living donors in the United States, compared with 12% in 1990.

Complications after transplantation are no greater in the ADPKD population than in the general population, and specific complications directly related to ADPKD are rare. Cyst infection is not increased after transplantation, and there is no significant increase in the incidence of symptomatic mitral valve prolapse or hepatic cyst infection. One study

showed an increased rate of diverticulosis and bowel perforation in ADPKD. Whether ADPKD increases the risk for development of new-onset diabetes mellitus after transplantation is controversial.

Although practiced routinely in the past, pretransplantation nephrectomy has fallen out of favor. By 1 and 3 years after kidney transplantation, kidney volumes decrease by 37.7% and 40.6%, whereas liver volumes increase by 8.6% and 21.4%, respectively.⁶⁵ Indications for nephrectomy include a history of infected cysts, frequent bleeding, severe hypertension, and massive kidney enlargement with extension into the pelvis. There is no evidence of an increased risk for development of kidney cell carcinoma in native ADPKD kidneys after transplantation. When nephrectomy is indicated, hand-assisted laparoscopic nephrectomy is associated with less intraoperative blood loss, less postoperative pain, and faster recovery compared with open nephrectomy and is increasingly being used.

SELF-ASSESSMENT QUESTIONS

1. Age of ESKD onset in the patient with ADPKD depends on:
 - A. mutated gene (*PKD1* vs. *PKD2*).
 - B. type of mutation.
 - C. modifier genes.
 - D. environmental factors.
 - E. all of the above.
2. Which one of the following statements about ADPKD and intracranial aneurysm is *true*?
 - A. All ADPKD patients should be evaluated for intracranial aneurysms.
 - B. Most intracranial aneurysms rupture.
 - C. The risk for rupture of an intracranial aneurysm depends on its size and location.
 - D. Magnetic resonance angiography to screen for intracranial aneurysms is contraindicated in advanced CKD because it requires the administration of gadolinium.
 - E. None of the above.
3. Which one of the following is the most common risk factor for nephrolithiasis in ADPKD?
 - A. Low urine citrate
 - B. Hypercalciuria
 - C. Hyperuricosuria
 - D. Hyperoxaluria
 - E. Renal tubular acidosis
4. Which of the following statements about ADPKD and hypertension is *true*?
 - A. ACE inhibitors have been shown to be superior to β -blockers to treat hypertension and protect kidney function in patients with ADPKD.
 - B. Rigorous BP control has been shown to be superior to standard BP control in prospective studies to protect kidney function in patients with ADPKD and CKD stage 3.
 - C. Both A and B are true.
 - D. Both A and B are false.

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